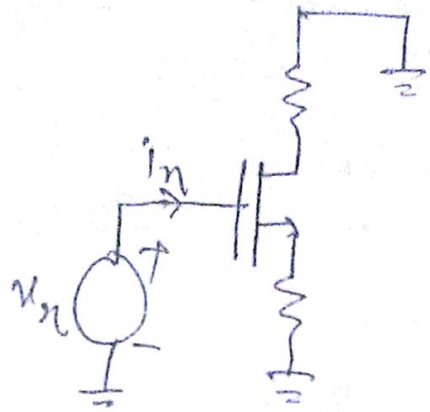
Lecture - 39

* I/O impedances of degenerated CS stage:-

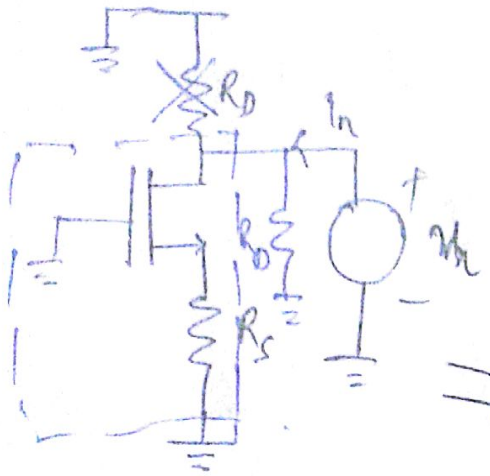
Input impedance:



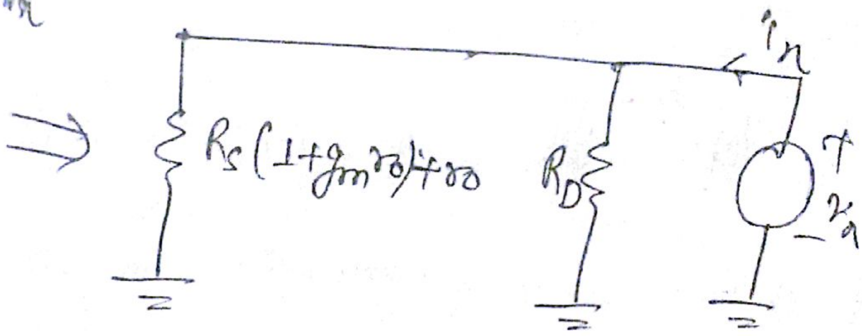
$R_{in} = \infty$

at low frequencies

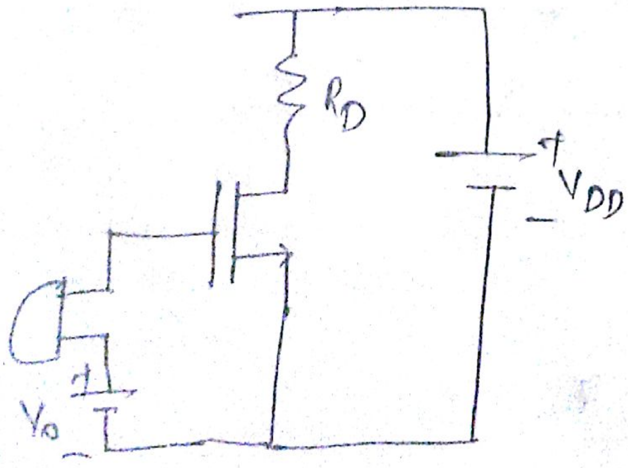
Output impedance: $\lambda > 0$

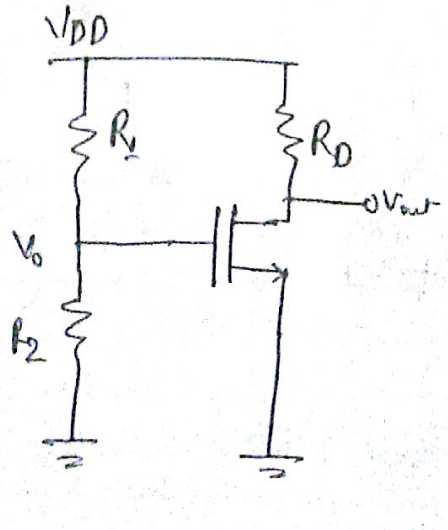


$$\boxed{R_{out} = R_D \,||\, \left[R_S(1+g_m r_o) + r_o\right]}$$



* Biasing Techniques:-

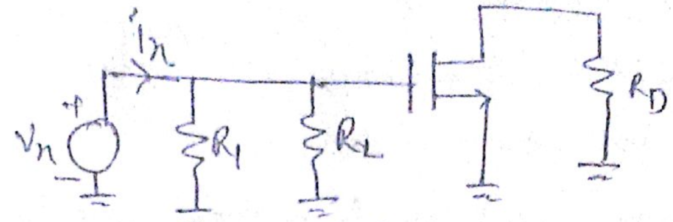




$$V_{DD} \times \frac{R_2}{R_1+R_2} = V_0$$

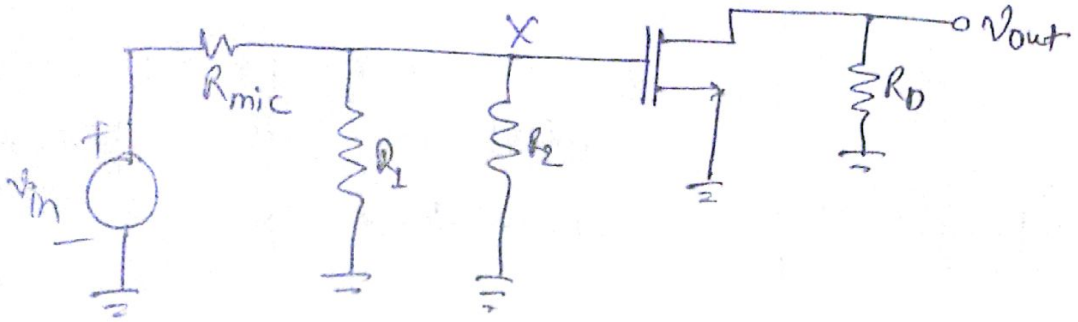
Observation:-

(1) Input impedance has fallen to $R_1 \| R_2$.



$R_{in} = R_1 \| R_2$

Connect a microphone which has some internal resistance.



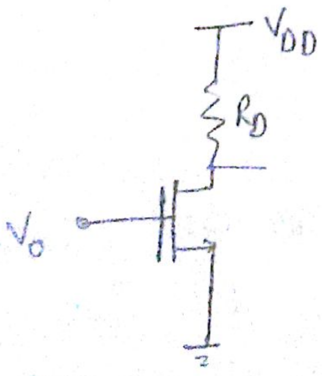
$\lambda = 0$

$$\frac{V_{out}}{V_{in}} = \frac{V_x}{V_{in}} \times \frac{V_{out}}{V_x} = \frac{R_1 \| R_2}{R_1 \| R_2 + R_{mic}} \times (-g_m R_D)$$

$\rightarrow$ We have an attenuation factor as a result of R_1, R_2 biasing components.

We choose $R_1 \| R_2 > R_{mic}$ to minimize the attenuation.

(2) Can we increase R_D to increase the gain?



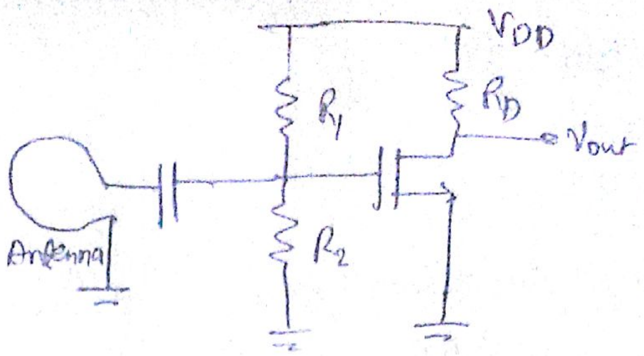
If R_D increases then V_D decreases therefore MOS goes closer to triode region. We can increase R_D only to the point where drain voltage is below the gate by one threshold voltage not more.

$$V_{DD} - I_D R_D \geq V_0 - V_{TH}$$

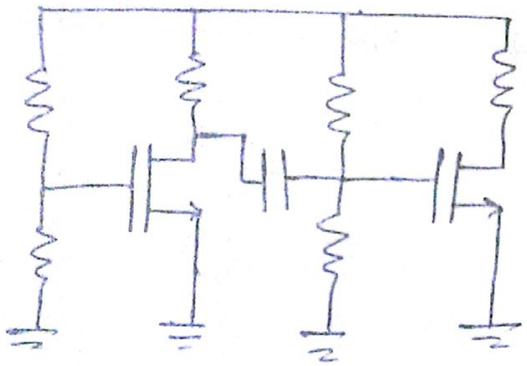
$$\frac{V_{DD} - V_0 + V_{TH}}{I_D} \geq R_D$$

(3)

→ This tells us that anytime we have an amplifier or any circuit that have some bias conditions and we want to



drive it with something (antenna, microphone or another amplifier) we have to worry about this agreement between the bias and conditions that the preceding stage produces.



* For matching the bias of both devices we place a capacitor between these stages.

(4)

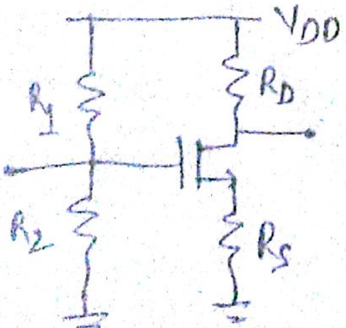
$$I_D = \frac{1}{2}\mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{Th})^2$$

$$= \frac{1}{2}\mu_n C_{ox} \frac{W}{L} \left(V_{DD}\frac{R_2}{R_1+R_2} - V_{Th}\right)^2$$

sensitive to : V_{DD}, process ($\mu_n C_{ox}$, V_{Th}), temperature

So generally the circuit is not so stable.

(5) Reduced sensitivity with degenerated CS stage



→ If the gate voltage is perturbs by some amount ΔV the this V_{GS} does not changes by ΔV it changes by less than ΔV and that means I_D is less sensitive to ΔV.

$$V_{DD}\frac{R_2}{R_1+R_2} = V_{GS} + I_D R_S$$

$$\because\ I_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS}-V_{TH})^2$$

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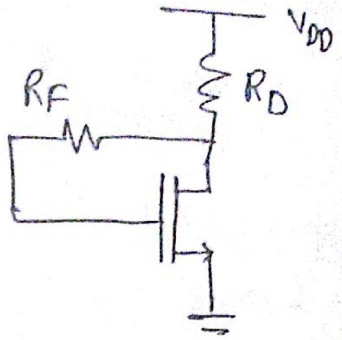
* Self bias CS stage :-



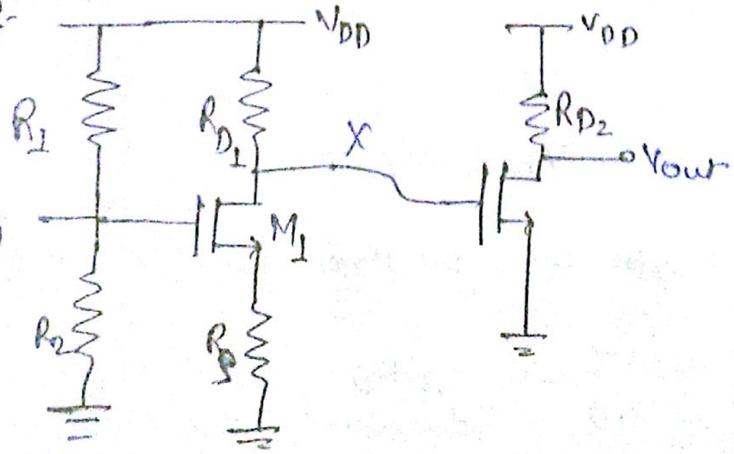
* If V_{TH} drops then I_D wants to increases, we draw larger current through R_D so drain voltage goes down. since no current goes through gate so gate voltage is equal to drain voltage therefore V_{GS} voltage goes down. Thus, I_D current goes down. Therefore this circuit is less sensitive to parameters. It has self correction mechanism.

$$\text{Drain voltage} = V_{DD} - I_D R_D = \text{Gate voltage}$$

$$\therefore\ I_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{DD}-I_D R_D - V_{TH})^2$$

* The device is always in ~~saturation~~ Saturation because the drain does not go below the gate so device is always in saturation.

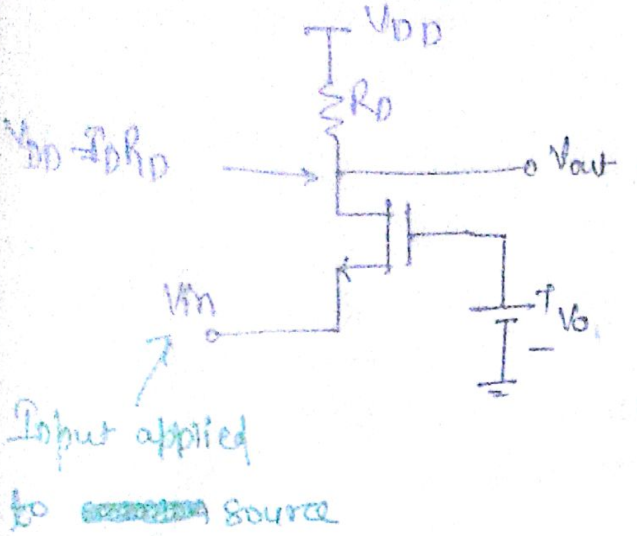
Example:-



$\lambda = 0$

$$\frac{V_{out}}{V_{in}} = \frac{V_x}{V_{in}} \times \frac{V_{out}}{V_x} = -\frac{R_{D1}}{\frac{1}{g_{m_1}}+R_S}\cdot(-g_{m_2}R_{D_2})$$

Common Gate Topology:-



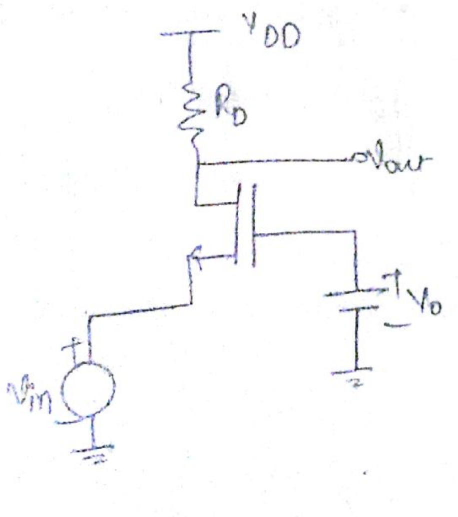
Step I:- Bias Calculation

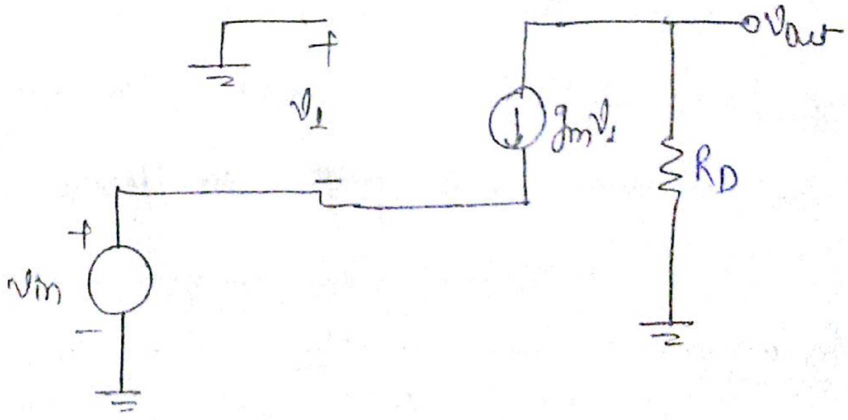
Step II: Small signal analysis

* Suppose source voltage goes up by ΔV and gate voltage is constant so V_{GS} decreases therefore I_D decreases so output goes up by some amount. Therefore this does not inverts i.e. it does not have negative sign in gain.

Lecture - 40

* Small signal properties :-



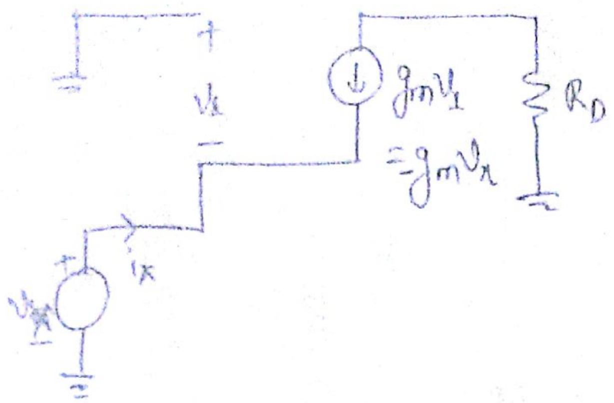


$$V_{in} = -V_1$$

$$V_{out} = -g_m V_1 R_D = -g_m(-V_{in}) R_D$$

$$\boxed{\frac{V_{out}}{V_{in}} = g_m R_D}$$

R_{in}:-

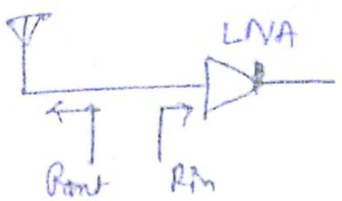


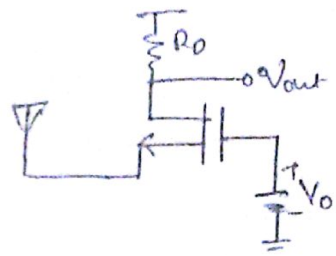
$$i_x = +g_m V_x$$

$$\boxed{R_{in} = \frac{V_x}{i_x} = \frac{1}{g_m}}$$

It's relatively very low.

Applications:-

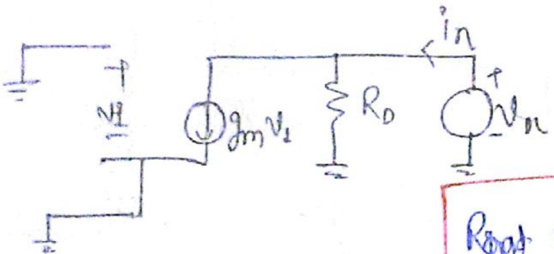




* we require input resistance of LNA be equal to the resistance of an antenna.

* The common gate matches impedances.

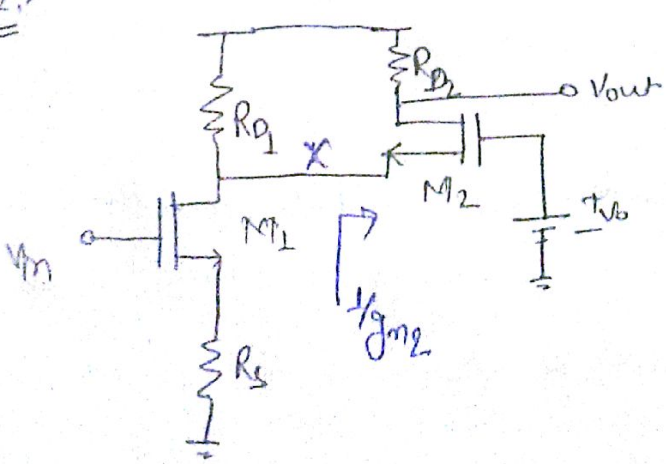
R_{out}:-



$V_1 = 0 \Rightarrow g_m V_1 = 0$

$$\boxed{R_{out} = \frac{V_x}{i_x} = R_D}$$

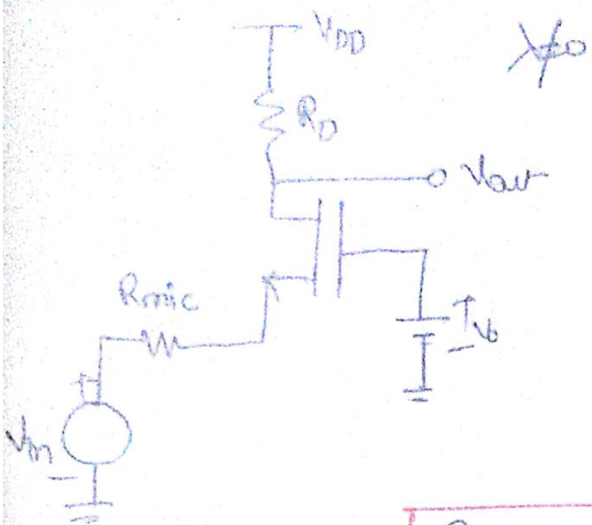
Example:-



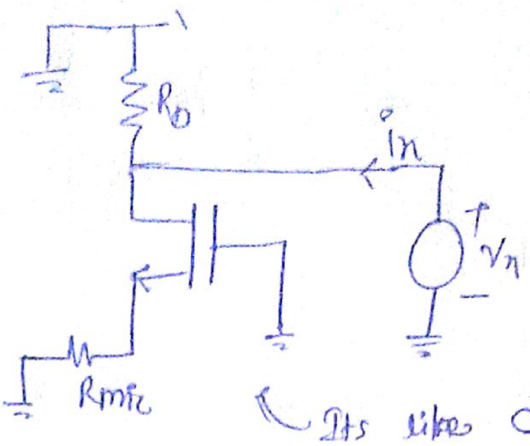
$\lambda = 0$

$$\boxed{\frac{V_{out}}{V_{in}} = \frac{V_x}{V_{in}} \times \frac{V_{out}}{V_x} = \left(-\frac{R_{D_1} \parallel \frac{1}{g_{m_2}}}{\frac{1}{g_{m_1}} + R_S}\right)\left(g_{m_2} R_{D_2}\right)}$$

Output resistance of CG stage in a special case:-



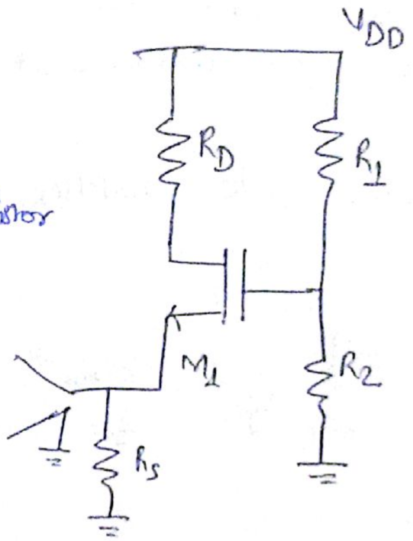
$\lambda \neq 0$



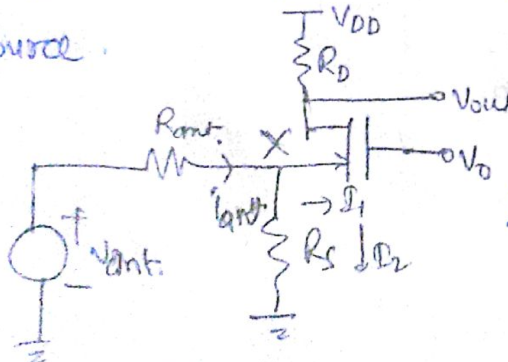
Its like CS stage with degeneration

$$\boxed{R_{out} = R_D \,||\, [(1+g_m r_o) R_s + r_o]}$$

Bias design:-



* The V.antenna does not allow dc.
* To bias properly, we just connect a resistor R_s with antenna.
* Since the antenna is not an ideal source.



$\lambda = 0$

When the current goes to divider then it splits up. When current go through R_s is undesirable. for reducing this current we just increase the value of resistance R_s.

$$i_1 = \frac{R_s}{R_s + 1/g_m}\, i_{ant}$$

$R_s >> 1/g_m$ for maximize the current i_1.

$$\frac{V_{out}}{V_{in}} = \frac{V_x}{V_{in}} \times \frac{V_{out}}{V_x} = \cancel{\left(\frac{R_S}{R_S + R_{ant}}\right)} \cancel{(g_m R_D)}$$

$$= \left(\frac{\left(\frac{1}{g_m} \| R_S\right)}{\left(\frac{1}{g_m} \| R_S\right) + R_{ant}}\right) \cdot (g_m R_D)$$

Example:

$V_{DD} = 1.5V$, $P = 1.5mW$, $\mu_n C_{ox} = 100\mu A/V^2$, $V_{TH} = 0.5V$, $\lambda = 0$

$\frac{W}{L} = 50$

Total current $= 1mA$

Assume $I_D \simeq 1mA$

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH})^2$$

$$\frac{2 \times 10^{-3}}{100 \times 10^{-6} \times 50} = (V_{GS} - V_{TH})^2$$

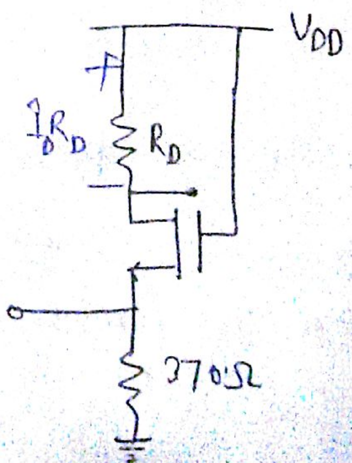
$\Rightarrow$ $V_{GS} = 0.63V + V_{Th} = 1.13V$

$$g_m = \frac{2 I_D}{V_{GS} - V_{TH}} = \frac{2 \times 10^{-3}}{0.13} = 3.2 mS = \frac{1}{316\,\Omega}$$

$$R_S = \frac{10}{g_m} = 3.16 k\Omega$$

we can't do that because if 1mA flows through it then voltage across it will be 3.16V which is not appropriate.

$$R_{S,max} = \frac{V_{DD} - 0.63}{1mA} = 370\,\Omega$$



$I_D R_D \leq V_{TH}$ (for M_1 be in saturation)

$$R_D \leq \frac{V_{TH}}{I_D}$$

$R_D = 500\,\Omega$ $\qquad A_V = g_m R_D = 1.6$

If we double $\frac{W}{L}$ for increasing gain,

$$\frac{W}{L} = 100$$

device becomes slower larger width and capacitance.

$$V_{GS} = 0.947V$$

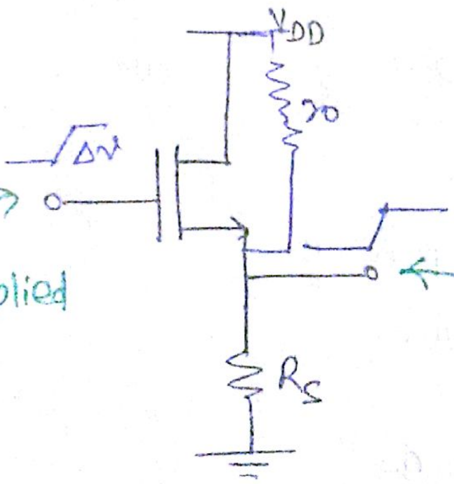
$$R_S = \frac{V_{DD} - V_{GS}}{I_D} = 550\,\Omega \qquad g_m = \frac{2I_D}{V_G - V_{th}} = 4.5mS = \frac{1}{220\Omega}$$

So current see 220Ω into the transistor so it flows through the transistor in large values.

$$\underline{g_m R_D = 2.3}$$

Lecture - 41

* Source Follower Topology:- (common drain)

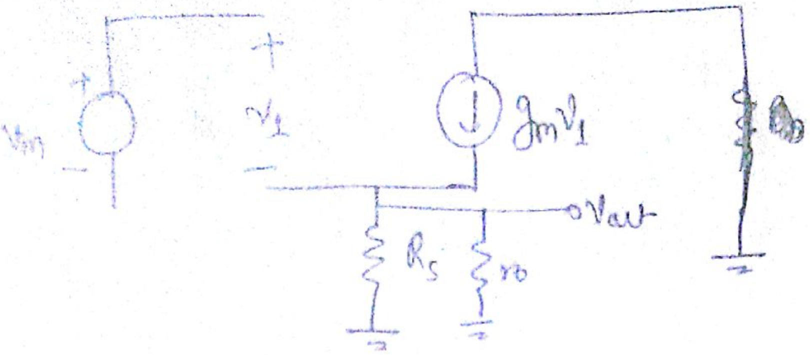


$\lambda = 0$

Input applied to gate

output sensed at source

* If we apply a change in voltage ΔV at input. V_{out} must also go up by some amount because if V_{GS} increases and V_S constant then I_D current increasing but drain current flows through R_S so it also goes up.



$V_{in} = V_1 + V_{out}$

$\Rightarrow V_1 = V_{in} - V_{out}$

$$g_m (V_{in} - V_{out}) = \frac{V_{out}}{R_s}$$

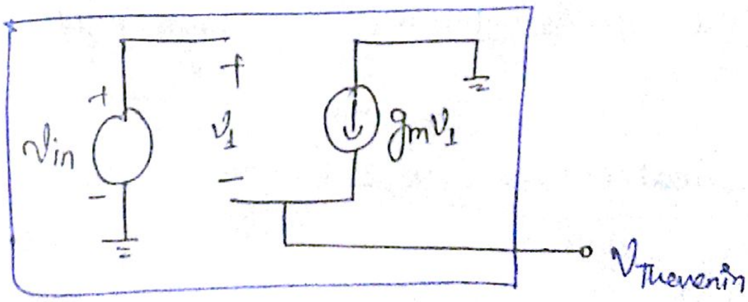
$$g_m V_{in} = V_{out}\left(\frac{1}{R_s} + g_m\right)$$

$$\Rightarrow \boxed{\frac{V_{out}}{V_{in}} = \frac{R_s}{\frac{1}{g_m} + R_s} = \frac{g_m R_s}{1 + g_m R_s}} < 1$$

$$\boxed{A_V = \frac{\text{resistance b/w source and ~~earth~~ ground}}{\frac{1}{g_m} + \text{resistance b/w source and ground}}}$$

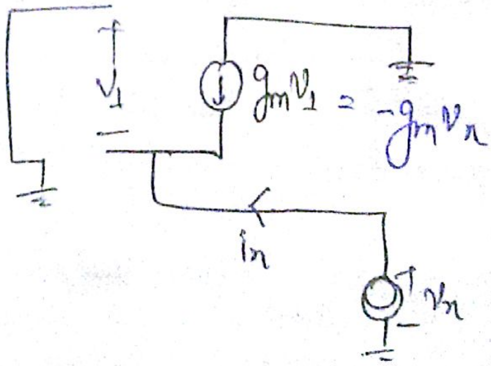
- Alternative Analysis:-

Thevenin equivalent →



$V_{in} = V_{Th}$

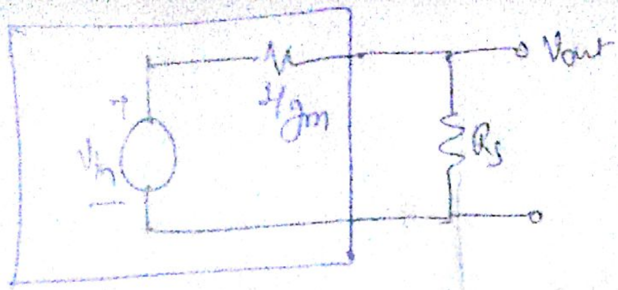
because $g_m V_1 = 0 \Rightarrow V_1 = 0$



$V_x = -V_1$

$\Rightarrow -g_m V_x + i_x = 0$

$\Rightarrow \frac{V_x}{i_x} = \frac{1}{g_m} = R_{Th}$ ← like a diode connected device.



This gives same result as previous approach.

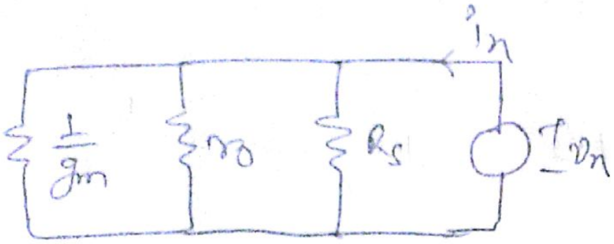
If $\lambda \neq 0$ then

$$A_v = \frac{R_S \| r_o}{R_S \| r_o + 1/g_m}$$

R_{in}:

$$\boxed{R_{in} = \infty \quad \text{At low frequency}}$$

R_{out}:

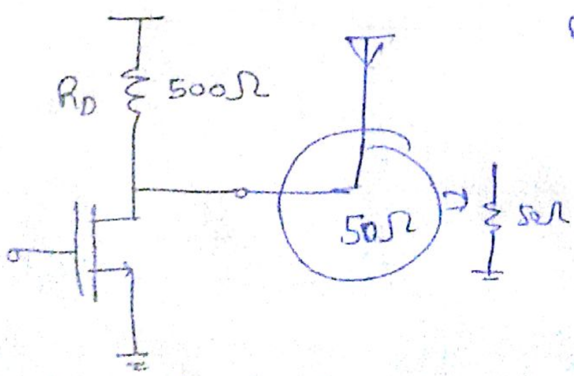


$$\boxed{R_{out} = \frac{V_n}{i_n} = \left(\frac{1}{g_m} \| r_o \| R_S\right)}$$

* The source follower has high input impedance and low output impedance.

Application of source follower:-

$\lambda = 0$



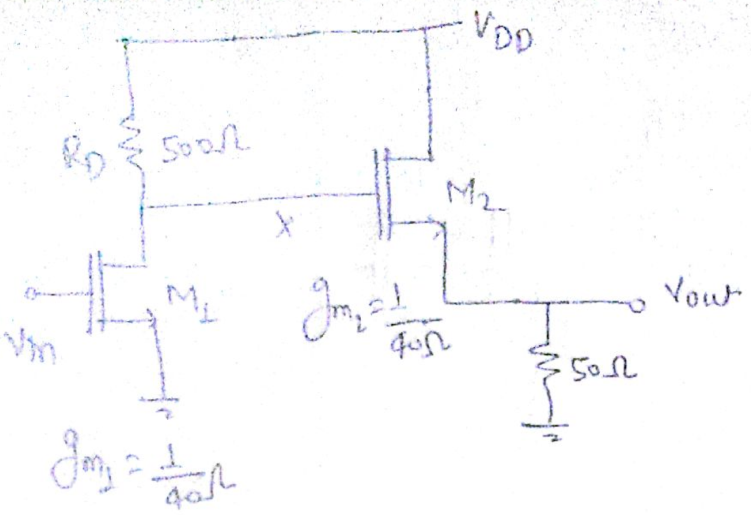
$g_m = \frac{1}{40\Omega}$

$A_v = -g_m R_D = -12.5$

* If we connect an antenna of 50Ω the R_D becomes is in parallel with 50Ω so A_v reduces

$$A_v = -\frac{1}{40\Omega}(500 \| 50) = -1.14$$

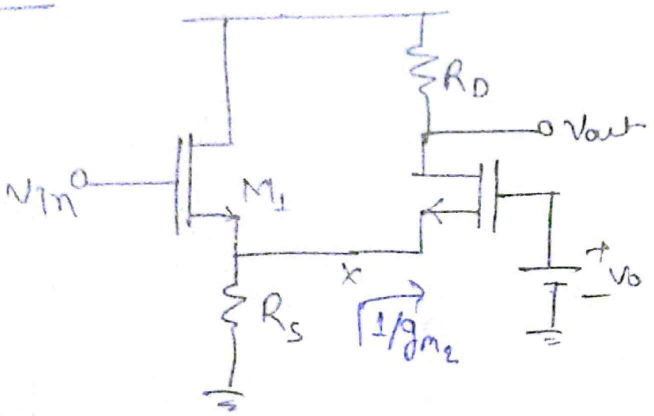
* We place a buffer to keep the gain of this device intact.



$$\frac{V_{out}}{V_{in}} = \frac{V_x}{V_{in}} \times \frac{V_{out}}{V_x} = (-12.5)\left(\frac{50}{50+40}\right) = -6.9$$

which is better than without buffer.

Example:



$$\frac{V_{out}}{V_{in}} = \frac{V_x}{V_{in}} \times \frac{V_{out}}{V_x} = \left[\frac{(R_s \| 1/g_{m2})}{(R_s \| 1/g_{m2}) + (1/g_{m1})}\right] \times \left[g_{m2} R_D\right]$$

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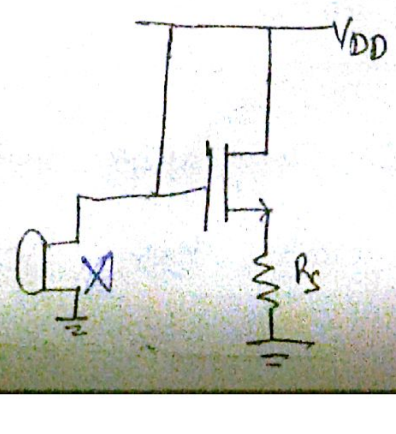
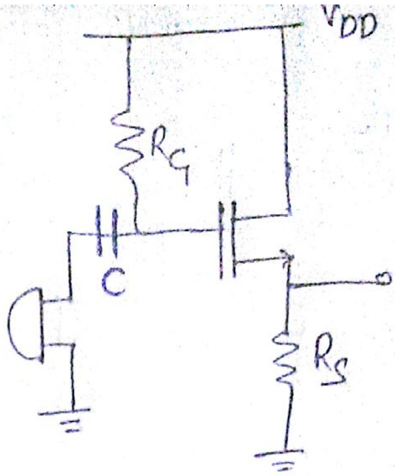
* **Bias design:-**

→ We cannot apply signal to the device because it is connected to the short ckt.

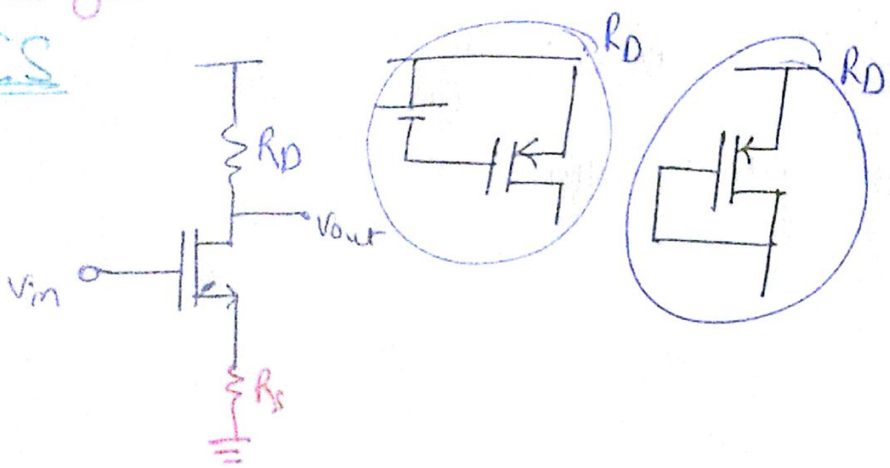


→ In some cases, the input device imposes its own dc value (like S.C), to avoid this we just connect a capacitor between them.



Summary :-

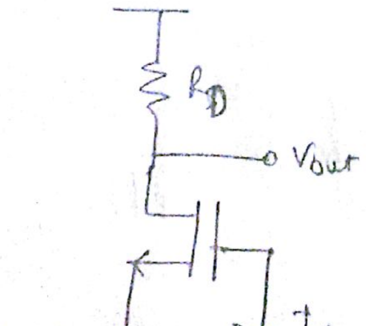
CS



$\lambda = 0$

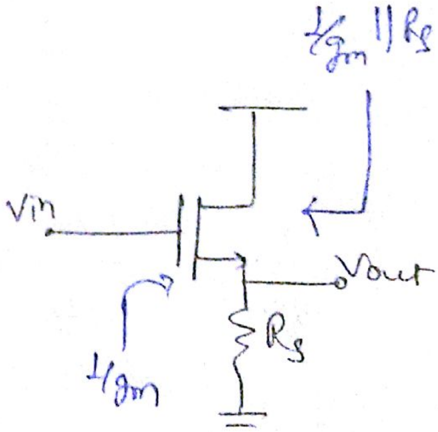
$$\boxed{A_v = -\frac{R_D}{R_S + 1/g_m}}$$

CG



$$\boxed{A_v = +g_m R_D}$$

SF



$$\boxed{A_v = \frac{R_S}{R_S + 1/g_m}}$$

* Because of its low output impedances (antenna, mic. etc) That's why we have use source follower as a buffer.